WEEK 3 · DAY 1 · RULE BOOK

Alarms That Cry Wolf

Follow it step by step. Every common error and its fix is here.

Your goal today: test a fixed alarm on satellite power telemetry, watch it cry wolf every eclipse night, fix it with context, and earn **14 green ✅ ticks**. Time: about 60 minutes. Keep this page open next to Colab.

Before you start (2 minutes)

- ☐ Open **W3D1-exercise** with the **Open in Colab** button.
- ☐ **File → Save a copy in Drive.** Work only in the new tab.
- ☐ Run **STEP 0** (Shift + Enter). Wait for ✅ `Ready!`

The 7 golden rules for today

#	Rule	Why
1	**An alarm is a True/False column**: `alarm = bus["col"] < 20`	Then `.sum()` counts how many times it fires
2	**Combine conditions with `&`, and wrap each one in brackets**	`(a < 20) & (b == 0)`. Python's `and` does not work on columns
3	`~` **means "not"**	`hot & ~after_eclipse` = hot, but not right after an eclipse
4	**The flag is a number, not text**: `== 0`, not `== "0"`	`eclipse_flag` holds 0 and 1
5	**Add context, don't widen the limit**	A wider limit also hides the real faults
6	**Always ask "where did that number come from?"**	A limit with no reason is a guess
7	**Rain is not a fault. Margin below 0 dB is.**	A detector must separate "raining" from "link down"

The exercise, step by step

Task	What you do	The tool you need
A1	Count eclipse hours	`bus["eclipse_flag"].sum()`
A2	Count the 20 A floor alarms	`floor_alarm.sum()`

Task	What you do	The tool you need
A3	How many of those are eclipse nights?	`(floor_alarm & (bus["eclipse_flag"] == 1)).sum()`
A4	Count the SoC < 82% alarms	`(bus["battery_soc_pct"] < 82).sum()`
A5	Which flag value means sunlit?	`0` or `1`
B1	Lower 3σ limit from sunlit hours	`sunlit.mean() - 3 * sunlit.std()`
B2	Sunlit hours below that limit	`(sunlit < low_limit).sum()`
B3	Temperature alarms right after eclipse	`(hot & after_eclipse).sum()`
C1	Rain alarms vs real outages	`cn_flags.sum()`, `(cn_flags & outage).sum()`
C2	Outages caught by a 2-hour rule	`two_in_a_row.sum()`
C3	Use uplink power as an input?	`"yes"` or `"no"`
C4	Name the three shapes	`"spike"`, `"level shift"`, `"drift"`
C5	Days below 95 Ah	`(bat["est_capacity_ah"] < 95).sum()`
🌶️ **Bonus**	Which date has two alarms?	read the A5 output

The alarm toolbox

```
alarm = bus["solar_array_current_a"] < 20                # a fixed limit = a True/False
column
alarm.sum()                                              # how many hours it fires
alarm & (bus["eclipse_flag"] == 0)                       # context: only when sunlit
bus["eclipse_flag"].shift(1) == 1                        # "the hour right after an
eclipse"
s.mean() - 3 * s.std()                                   # a 3-sigma limit from the
data
outage & outage.shift(1, fill_value=False)               # "bad for 2 hours in a row"
bus.index[alarm].tolist()                                # WHEN did it fire?
```

When things go wrong: error → fix

You see	It means	Fix
`ValueError: The truth value of a Series is ambiguous`	You used `and` / `or` between columns	Use `&` / `|` and brackets: `(a < 20) & (b == 0)`

You see	It means	Fix
`TypeError: unsupported operand type(s) for &`	Missing brackets around a comparison	`(bus["eclipse_flag"] == 0)` in brackets
A count of `0` that should not be 0	Compared a number with text (`== "0"`)	Use `== 0`
`NameError: name 'floor_alarm' is not defined`	You skipped A2, or restarted	Run STEP 0 and the cells above
`AssertionError: ✍️ Fill in SUNLIT first`	A5 still has `___`	Put `0` or `1`
`KeyError: 'eclipse_flag'`	Typo in the column name	Copy it exactly from `bus.head()`
C1 says wrong format	Gave one number instead of two	`check("C1", (n_flags, n_real))`
Everything is weird	–	**Runtime → Restart session and run all**

Stuck? Climb the ladder

1. **Read the ❌ hint.**
2. **Check this rule book's error table.**
3. **Ask your neighbour.**
4. **Ask the AI button** to explain the error. Course data only, never real telemetry.
5. **Raise your hand.**

🏁 **Finished?** Run `scoreboard()` . 14 of 14 = 🏆 . Then try the 🌶️ bonus, and think: which alarm in *your* unit cries wolf?